Yunus Serhat Bıçakçı, PhD

yunusserhat@gmail.com • yunusserhat.com •
GitHub: yunusserhat • LinkedIn: yunusserhat
London, UK & Istanbul, Türkiye

Research interests

Data Science, Machine Learning, Deep Learning, Computer Vision, Generative AI, Geospatial, GeoAI

Positions

2025 — Present

Assistant Professor
Marmara University, Faculty of Applied Sciences, Department of Artificial Intelligence and Machine Learning – Istanbul, Türkiye
Teach undergraduate technology courses, developing syllabi and delivering engaging lectures that enhance student understanding of concepts. Continue active research, focusing on data science and geospatial, with findings published in high-impact journals and presented at international conferences. Serve on academic committees, contributing to curriculum development and the strategic planning of departmental goals. Conducted regular office hours to provide academic support and career advice to students.

2024 – Present

Research Software Engineering Scientist
University of Glasgow, School of Geographical and Earth Sciences, Geospatial Data Science Group – Glasgow, UK
Initially appointed as Research Associate and subsequently continuing as Affiliate Research Software Engineering Scientist within the Geospatial Data Science Group. Contribute to collaborative projects on geospatial data science and multimodal AI, co-author scholarly publications, and present findings at international conferences. Support proposal development and mentoring, maintaining active collaboration with the group on ongoing research initiatives.

2023 — 2024

Postdoctoral Research
University College London (UCL), Jill Dando Institute of Security and Crime Science – London, UK
Conducted research on AI-driven crime mapping, developing ML models (NLP, geospatial AI, deep learning) for large-scale data analysis. Authored and published scholarly papers, contributing to the knowledge base in the field of data science and environmental criminology. Presented research findings at international conferences and seminars, engaging with academic peers and industry experts.

2013 — 2023

Research Assistant
Marmara University, Faculty of Applied Sciences, Department of Artificial Intelligence and Machine Learning – Istanbul, Türkiye
Conducted comprehensive research for academic projects, specialising in geospatial information technology, contributing significantly to the data collection and analysis phases. Assisted in the publication of several peer-reviewed articles. Supported the Head of School in administrative duties, streamlining daily operations and enhancing organizational efficiency within the department.

Education

2014 – 2021 **Istanbul Technical University** – Istanbul, Türkiye
PhD in Applied Informatics
Geographical Information Technologies Program
Supervisor: Prof Dr Dursun Zafer Şeker

2011 – 2013 **Sakarya University** – Sakarya, Türkiye
MSc in Informations Systems
Management Information Systems Program
Supervisor: Prof Dr Orhan Torkul

2007 – 2011 **Anadolu University** – Eskişehir, Türkiye
BA in Business Administration

2007 – 2011 **Ankara University** – Ankara, Türkiye
BSc in Engineering

Book Chapters

2026 **Towards GeoAI Foundation Models: A Multimodal Learning Framework with Spatial Knowledge**
Meiliu Wu, Qunyin Huang, Anahid Basiri, Yunus Serhat Bıçakçı, Joseph Shingleton.
In Janowicz, K. et al. (eds), Geography according to ChatGPT, Frontiers in Artificial Intelligence and Applications, IOS Press, forthcoming (Accepted).

2025 **Big Data and Social Media Analytics: Opportunities for Interactive Crime Mapping**
Yunus Serhat Bıçakçı, Alina Ristea, Kate Bowers.
New Research in Crime Modeling and Mapping Using Geospatial Technologies, In: Leitner, M. (eds), New Research in Crime Modeling and Mapping Using Geospatial Technologies. Geotechnologies and the Environment, Springer, Dordrecht. *(DOI)*

Towards GeoAI Foundation Models: A Multimodal Learning Framework with Spatial Knowledge
Meiliu Wu, Qunyin Huang, Anahid Basiri, Yunus Serhat Bıçakçı, Joseph Shingleton.
In Janowicz, K. et al. (eds), Geography according to ChatGPT, Frontiers in Artificial Intelligence and Applications, IOS Press, forthcoming (Accepted).

2022 **Artificial Intelligence Applications in Tourism and Travel Offices**
Yunus Serhat Bıçakçı, Bora Aksu
Digital Transformation Dynamics in Tourism, In Semiz Celik, D. (eds), The Reflections of Digital Transformation on Tourism (pp. 139-152). Istanbul, Turkey: Beta Publishing.

Publications

2026 **Performance Comparison of Multimodal Vision-Language Models in Classifying Turkish Dishes** *(DOI)*
Yunus Serhat Bıçakçı.
Sakarya University Journal of Computer and Information Sciences, 9(1), 119–133.

Emergent Spatio-Semantic Structure in Large Language Model Embedding Spaces *(DOI)*
Joseph Shingleton, Yunus Serhat Bıçakçı, Yu Wang, Ana Basiri.
EarthArXiv, 2026.

Evaluating Explanation Integrity in a Vision Language Model for Urban Perception
Joseph Shingleton, Yunus Serhat Bıçakçı, Meiliu Wu, Yu Wang, Ana Basiri.
Transactions in GIS (Special Issue: Ethical and Explainable GeoAI), under review, 2026.

Hybrid CNN-Transformer Ensemble for Enhanced Tank Detection in Aerial Imagery *(DOI)*
Yunus Serhat Bıçakçı.
Research Square preprint, 05 February 2026; The Visual Computer (Research article), in peer review, 2026.

2025 **Street-Level Geolocalization Using Multimodal Large Language Models and Retrieval-Augmented Generation** *(DOI)*
Yunus Serhat Bicakci, Joseph Shingleton, Anahid Basiri.
arXiv:2509.01341 [cs.CV]

CNN and Transformer U-Nets in Multiple Sclerosis Lesion Segmentation: A Comparative Assessment *(DOI)*
Beytullah Sarıca, Yunus Serhat Bıçakçı, Dursun Zafer Şeker.
International Journal of Imaging Systems and Technology (IMA)

2023 **ATTransUNet: Semantic Segmentation Model for Building Segmentation from Aerial Image and Laser Data** *(DOI)*
Yunus Serhat Bıçakçı, Beytullah Sarıca.
Nordic Machine Intelligence, 2(3).

2020 **Location-Based Analyses for Electronic Monitoring of Parolees** *(DOI)*
Yunus Serhat Bıçakçı, Dursun Zafer Seker, Hande Demirel.
ISPRS International Journal of Geo-Information, 9(5), 296.

2019 **Spora Yönelik Tutum İle Yaşam Doyumu Arasındaki İlişki: Marmara Üniversitesi Sosyal Bilimler Meslek Yüksekokulu Öğrencileri Örneği /in Turkish** *(Link)*
Cağdaş Caz, Yunus Serhat Bıçakçı, Fatma Firuzan Nakipoglu.
Gümüşhane Üniversitesi Sosyal Bilimler Enstitüsü Elektronik Dergisi, 10(3), 484–491.

2017 **Spatio-temporal analyses to estimate speed information for transport forecasting and scenario testing** *(Link)*
Yunus Serhat Bıçakçı, Beytullah Sarıca, Mete Ercan Pakdil, Batuhan Yazırlı, Hande Demirel.
Fresenius Environmental Bulletin, 26(1).

Software & Repositories

2026 **GeoAI-VLM: Geospatial Vision-Language Model Analysis**
Yunus Serhat Bıçakçı.
GitHub repository; Zenodo archived release v0.2.1 (DOI/Release).

sppt: Spatial Point Pattern Test for Aggregated Data (Python)
Yunus Serhat Bıçakçı.
GitHub repository; Zenodo archived release v0.1.7 (DOI/Release).

Presentations

2026 **Mapping the Semantics of the Street: A VLM-Driven Geospatial Analysis in Fatih, Istanbul**
Yunus Serhat Bıçakçı, Joseph Shingleton, Yu Wang, Anahid Basiri.
1st International Conference on Geospatial Artificial Intelligence (GeoAI 2026), Ghent, Belgium, 3–6 June 2026, oral presentation accepted.

Emergent Spatio-Semantic Structure in Large Language Model Embedding Spaces
Joseph Shingleton, Yunus Serhat Bıçakçı, Yu Wang, Ana Basiri.
1st International Conference on Geospatial Artificial Intelligence (GeoAI 2026), Ghent, Belgium, 3–6 June 2026, oral presentation accepted.

The Invisible City: Not All Streets Are Seen Equally
Yu Wang, Joseph Shingleton, Yunus Serhat Bıçakçı, Ana Basiri, James Ackland.
1st International Conference on Geospatial Artificial Intelligence (GeoAI 2026), Ghent, Belgium, 3–6 June 2026, oral presentation accepted.

A graph-based community detection approach for identifying the semantic neighbourhoods within London's Airbnb properties
Joseph Shingleton, Yunus Serhat Bıçakçı, Yu Wang, Ana Basiri, Meiliu Wu.
29th AGILE Conference on Geographic Information Science, Smart Data: Making Geospatial Data Actionable, Tartu, Estonia, 16–19 June 2026, short paper accepted.

2025 **Multimodal Semantic Segmentation for Impervious Surface Mapping**
Yunus Serhat Bıçakçı.
15th Workshop on Hyperspectral Image and Signal Processing: Evolution of Remote Sensing (WHISPERS 2025), 12-14 November 2025, Bellaterra (Barcelona), Spain.
From Video to Recipe: Leveraging Vision Language Models for Culinary Analysis
Yunus Serhat Bıçakçı.
4th International Congress on Food Researches, 16-18 October 2025, Sivas, Türkiye.

2023 **Using Big Data, Artificial Intelligence and Cloud Technologies for Interactive Crime Mapping**
Yunus Serhat Bıçakçı, Alina Ristea, Kate Bowers.
Stockholm Criminology Symposium. (Link)

Application of Natural Language Processing Techniques for Crime Detection
Yunus Serhat Bıçakçı, Alina Ristea, Kate Bowers.
Environmental Criminology and Crime Analysis (ECCA) 2023 (Link).

2022 **Artificial Intelligence Applications for Social Municipality: The Case of Chatbot**
Yunus Serhat Bıçakçı.
International VI. Civil Strategy Symposium. (Link)

Projects

2026 – 2027 **AI-Based Analysis of Istanbul Street-Level Imagery and Open Dataset Creation**
Research Universities Support Program (Project ID/Code: 12127; Duration: 12 months; Budget: 546,500.00 TL; Principal Investigator: Dr. Yunus Serhat Bıçakçı)
Scope: Large-scale collection of approximately 3 million street-level images across Istanbul from crowdsourced and open platforms (e.g., Mapillary), AI-based analysis of the collected imagery, and publication of both imagery and derived attributes as an open dataset.
Method: a four-stage pipeline including (i) data collection and preprocessing, (ii) AI-driven analysis, (iii) quality assessment and validation through a reference subset, and (iv) open public release of the dataset.
Originality and expected impact: beyond data collection, the project systematically applies and evaluates state-of-the-art AI methods on Istanbul-scale street imagery and disseminates outputs as open research assets, enabling interdisciplinary studies in urban morphology, sociology, security services, urban planning, computer vision, AI/ML, and quantitative social sciences, while aligning with national priorities on AI, smart cities, and open data.

2025 **Multimodal Semantic Segmentation for Impervious Surface Mapping**
Developed multiple deep learning models for the IEEE WHISPERS 2025 ISASeg Challenge, focusing on impervious surface segmentation using Sentinel-1 and Sentinel-2 imagery. Explored and fine-tuned various architectures including DeepLabV3+, UPerNet, SegFormer, and U-Net with ensemble methods. Ranked 1st in both Phase 1 and Phase 2 leaderboard, achieving top segmentation scores through multimodal data fusion (Phase 1 Score: 0.8542; Phase 2 Score: 0.8241). This work demonstrated the effectiveness of combining synthetic aperture radar (SAR) and multispectral optical data for robust urban mapping tasks, addressing challenges such as varying spatial resolutions, noise, and seasonal variability. The project not only advanced state-of-the-art methods in semantic segmentation but also provided insights into practical applications for urban planning, climate resilience monitoring, and sustainable infrastructure development.

Awards, and Short-Term Study

2024 - Present **Global Talent Endorsement – The British Academy**
Received a prestigious endorsement as part of the UK's Global Talent visa scheme, recognizing exceptional promise contributions. This endorsement facilitates ongoing academic and research pursuits in the UK, highlighting significant impact in my speciality.

2023 - 2024 **TUBITAK Postdoctoral Fellowship**
Awarded by the Scientific and Technological Research Council of Türkiye (TUBITAK) for postdoctoral research at University College London (UCL). This fellowship provided GBP 24,600 in financial support, in addition to travel expenses, recognizing outstanding research potential.

2022 **Google Machine Learning Bootcamp** – *Certificate of Achievement*
Study time: 6 months

Workshops

2024 **15th International Crime Science Conference**. (Online) 10 June, 2024, London, UK.

2023 **Mobile Phone Data for Mobility Workshop**. (Online) 06-07 December, 2023, UK.

Spatial Data Science Across Languages (SDSL) . (In-person) 18-19 September, 2023, Münster, Germany.

Scottish Crime Data Hackathon. (Online) 25th July, 2023, Edinburgh, Scotland.

14th International Crime Science Conference. (Online) 19 June, 2023, London, UK.

2nd Cardiff NLP Summer Workshop. (In-person) 26-27 June, 2023, Cardiff, UK.

Teaching Experience

2021 – Present **Instructor, Marmara University**
Information and Communication Technologies (BSP1005)
Management Information Systems (ILT2006 / BYS2018)
Office Applications (BSP1060 / BSP1071 / BSP2007)
Media Literacy (ILT2005)
Telephone Communication and Time Management (BYS2032 / BYS2034)
Fundamentals of Computing (BSP205)
Delivered lectures and practical sessions, developed course materials, and supervised student projects.

Doctoral Dissertation

2021 **"Spatio-Temporal Analysis of Applied Electronic Monitoring on Parolees of Mobility: A Case Study in Istanbul"** Istanbul Technical University *(Link)*

Mentorship

2022 – Present **Mentor**
inzva (Non-Profit Organization)
Mentored career mentees in artificial intelligence programs, facilitated skill development, and supported participants in their learning paths.

University Service

2022 – 2023 **Editorial Board (Co-Editor)**
The Social Research Journal of Marmara University

2021 – 2022 **Journal Referee**
International Humanities and Social Science Review (IHSSR)

2013 – 2022 **Marmara University, Vocational School of Social Sciences**
Member of Board, Department of Secretary and Office Management
Academic Process Manager, Department of Secretary and Office Management
Web Coordinator, Vocational School of Social Sciences (2014 – 2018)

Talks and Certifications

1st Place Award (Multimodal Segmentation Challenge) — *Certificate*

TensorFlow Developer — 10/2022–10/2025 *Certificate*

Hugging Face Agents Course — 06/2025 *Certificate*

Deep Learning Specialization (Coursera) *Cred.*
TensorFlow Developer (DeepLearning.AI) *Cred.*
NVIDIA: Fundamentals of Deep Learning *Certificate*

Intro to ML (Talk) – Google DSC Işık University — 11/2022 *Video*
TensorFlow Basics (Talk) – Google DSC Bandırma University — 12/2022 *Video*
Google Cloud Skills Boost *Profile* — **Google Developer** *Page*

DataCamp: Intermediate R *Certificate*

Skills

Programming – Python, R, SQL

Geospatial Stack — ArcGIS, QGIS, GeoPandas, PostGIS, DuckDB

Generative AI & Inference — Transformers, vLLM, LMDeploy, Ollama, OpenAI API, quantisation (AWQ, GPTQ)

RAG & Agentic Systems — MCP, LlamaIndex, LangGraph, FAISS, vector databases

Computer Vision — ViT-family (CLIP, SigLIP, DINO), semantic segmentation, object detection

Multimodal — VLMs, evaluation, dataset curation

MLOps & Deployment — Docker, CI/CD, model serving, monitoring

Data & Machine Learning — OpenCV, NumPy, Pandas, scikit-learn

Cloud & Tools — Google Cloud Platform, Microsoft Azure, Git